From Directional Statements to Operationalizable Indicators

——Operationalization Pathways and Proof-of-Concept for the Three Criteria of Complexity Metabolic Failure

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Theoretical Affiliation

This paper belongs to the Civilization Dynamics research program.

Theoretical Foundations: *Systems Ontology*; *Nooecology*

Methodological Foundations: The Criterion Dual-Reversal Method and the Boundary Anchoring Method. The former generates criteria; the latter deploys them. Together, they constitute the methodological pathway from grand theory to empirical tractability.

Methodological Statement

What this paper executes is a proof of concept, not an empirical test.

Argumentative Strategy: Taking the three criteria of Complexity Metabolic Failure as a case instance, directional statements are transformed into operationalizable indicators through logical reversals, demonstrating the actual operation of the Criterion Dual-Reversal Method.

Pre-Declaration: This paper does not claim that "failure is currently occurring." It does not calibrate failure thresholds. It does not perform cross-criterion coupling and integration. These are tasks for subsequent work; their absence is not a flaw intrinsic to this paper.

The contribution of this paper lies in demonstrating that the pathway exists and is traversable.

Abstract

Grand social theory has long confronted a methodological predicament: its core variables—class consciousness, social solidarity, the dispersion of power, meta-cognitive capacity—are rigorously coherent within their theoretical architectures yet remain stubbornly resistant to direct measurement and operationalization. The theory of Complexity Metabolic Failure (CMF) inherits this very predicament. The theory proposes three criteria—Diminishing Returns of Coordination, Response Latency Deadlock, and Antagonism Failure—for diagnosing whether a system is sliding toward metabolic failure. Yet in their existing formulation, these three criteria remain at the level of directional statements: they indicate the direction in which to observe, but they supply no concretely executable measurement protocol.

The task of this paper is not to subject CMF theory to empirical test. It is, rather, to carry out a proof of concept: to demonstrate that these three criteria can be transformed from directional statements into operationalizable indicators, and that this transformation pathway is traversable under the resource constraints of an independent researcher.

This paper adopts logical reversal as its core operational method: when a grand-theoretical variable cannot itself be measured directly, one turns instead to measuring the system-level consequences produced by the absence of that variable. The transformation of each criterion follows a four-step logical chain—original predicament → key reversal → preconditions → operationalization result—thus exhibiting the complete operation of the Criterion Dual-Reversal Method. Taking the construction industry as the single-sector proving ground and 2000–2020 as the unified temporal window, the paper completes an operationalization protocol for each of the three criteria in sequence: the revision frequency of national standards proxies for "runaway rule-revision velocity" (Criterion A); a three-anchor timeline—authoritative hazard designation → first paradigmatic publicly-reported accident → issuance of targeted standard—measures "rule-response latency" (Criterion B); and generational rupture among construction workers, the attenuation of the purchasing-power ratio, and the discursive pivot in official media together track "the declining explanatory coverage of the meta-narrative" (Criterion C).

All three operationalizations succeeded in obtaining openly-available official empirical data, and the data are broadly consistent with the theoretically-expected direction of each criterion. The paper does not, on this basis, assert that "failure is currently occurring." It asserts only the following: grand-theoretical variables, however amorphous, can be transformed into observable signals through logical reversal; and the Criterion Dual-Reversal Method as a methodological instrument has, in the concrete case of the CMF criteria, traversed the complete pathway from "directional statement" to "operationalizable indicator."

This paper is the second in the verification sequence of the Boundary Anchoring Method. Together with the already-completed "involution verification," the two constitute the complete methodological chain from grand theory to empirical tractability. The involution verification demonstrated "what the method is"—the sorting capacity of the Boundary Anchoring Method when deploying already-calibrated criteria. This paper demonstrates "where the criteria come from"—the operationalization feasibility of the Criterion Dual-Reversal Method when generating criteria. Taken together, they open a traceable, replicable, and independently testable empirical pathway for all grand-theoretical variables—not for Complexity Metabolic Failure alone.

Keywords : Complexity Metabolic Failure; criterion operationalization; logical reversal; Criterion Dual-Reversal Method; proof of concept; grand theory

Concept Preview and Reading Guide

The argumentative structure of this paper differs from that of a conventional empirical study. It does not begin from a literature review and conclude with hypothesis testing. Its core is a methodological demonstration—transforming the amorphous variables of grand theory from directional statements into operationalizable indicators, and in the process, publicly exhibiting every step of the logical reversal involved. The following remarks aim to equip the reader with an appropriate set of expectations and a preliminary familiarity with the paper's core terminology.

I. Operational Concepts Developed in This Paper

The following concepts represent the original methodological contributions of this paper. Each is accompanied by a concise operational definition.

Concept	Operational Definition
Criterion Operationalization	The process of transforming a directional criterion statement (e.g., "the growth rate of coordination costs persistently exceeds the growth rate of systemic utility") into proxy variables and analytical procedures measurable with publicly available data
Proxy Variable	When the core variable of a criterion cannot itself be directly measured, an observable substitute indicator that carries information about the system-level consequences of that variable's absence. Example: using "standard revision frequency" to proxy for "coordination cost growth rate"
Logical Reversal Chain	The four-step reasoning structure of criterion operationalization: Original Predicament (why it cannot be measured directly) → Key Reversal (where to pivot the thinking) → Preconditions (the presuppositions on which the reversal's validity rests) → Operationalization Result (the observable signal ultimately locked in)
Proof of Concept	The argumentative positioning of this paper. It does not assert the truth of the criteria; it does not declare that failure is currently occurring. It demonstrates only that the transformation pathway exists and is traversable under specified constraints

II. Concepts Imported from the Parent Theoretical Framework

The following concepts originate in the theory of Complexity Metabolic Failure and the Criterion Dual-Reversal Method. This paper does not presuppose that the reader is conversant with the parent framework; each term is supplied here with only the minimal operational definition required for this paper's purposes.

Concept	Minimal Definition as Used in This Paper	Full Statement
Complexity Metabolic Failure (CMF)	A state of functional congestion into which a system enters when the incremental information produced by the lateral interactions of its subsystems persistently exceeds the sense-making—analysis—response bandwidth of its top-level governance protocols. The system continues to receive effective input yet produces disordered output	See the CMF paper
Criterion A (Diminishing Returns of Coordination)	The efficiency-layer criterion of CMF: the growth rate of macro-coordination costs persistently exceeds the growth rate of total systemic utility	Ibid.
Criterion B (Response Latency Deadlock)	The temporal-layer criterion of CMF: the system's typical response latency to critical interaction events persistently exceeds the crisis diffusion speed of those events	Ibid.
Criterion C (Antagonism Failure)	The meaning-layer criterion of CMF: the explanatory coverage of the system's top-level meta-narrative exhibits irreversible attenuation over intergenerational observation cycles	Ibid.
Criterion Dual-Reversal Method	The methodological instrument responsible for generating criteria. A two-step operation: ① deriving observable criterion signals from the dynamic inequality of the pathological state; ② reversing the direction of that inequality to obtain the set of criteria for the healthy state	See the Methodological Appendix of *Nooecology*
Boundary Anchoring Method	The methodological instrument responsible for deploying criteria. Taking an already-calibrated decay limit as the anchor, it calculates the deviation degree of the system under test relative to that anchor	Ibid.; for its first independent verification, see the involution verification paper

III. Easily Confused Concepts Requiring Prior Clarification

Concept	Meaning in This Paper	Is NOT...
Metabolic	The information-processing and coordination process performed by the system's top-level governance	Physiological or everyday "metabolism"

Concept	Meaning in This Paper	Is NOT...
	protocols upon the interaction effects of subsystems, comprising three links: perception, analysis, and response	
Failure	An independent dynamic phase in which the system is still operating but has entered functional congestion—input remains effective, output becomes disordered. Distinguished from collapse	The cessation or disintegration of the system; exhaustion from which one recovers
Criterion	An observable signal used to judge whether a system is approaching a specified state	An index, an indicator, or a complete measurement instrument
Reversal	The logical operation of displacing the measurement object from "the variable itself" to "the consequences of the variable's absence"	A change of opinion, a swing of stance, or the dialectical "thesis-antithesis-synthesis"

IV. Special Note on Genre and Structure

Why does this paper lack a conventional literature review?

A conventional paper anchors its position in the knowledge landscape through a front-loaded literature review. The starting point of this paper is not an internal gap within the existing literature, but an operationalization lacuna that has not yet been named. Forcing a literature review into the front matter would generate a misalignment of argumentative pathways. Dialogue with existing methodologies is instead embedded in Chapter 2 (Methodological Background) and Chapter 6 (Discussion and Boundary Statements).

Does the argument of this paper depend on prior acquaintance with the parent theory?

Not entirely. All concepts imported from the parent framework are supplied with dedicated operational definitions in the table above. The argument proceeds from these operational definitions and is logically self-contained. Footnote references to the theoretical parent framework serve as deep-reading interfaces; they do not constitute necessary premises for following the core argument of this paper.

Does the density of new concepts amount to academic jargon?

The number of terms introduced in this paper is proportionate to the structural character of the concepts themselves. To describe a novel methodological operation with precision, a supporting set of terms is needed to fix the exact boundaries of each step. To prevent these terms from degenerating into jargon, this paper adopts three defensive measures: the concentrated preview provided in this section; an operational explanation in a standalone paragraph the first time each concept appears in the main text; and a worked example in Chapter 3 (the transformation process of the three CMF criteria) that gives abstract terms a concrete anchor.

V. Suggested Reading Pathways

- **For rapid comprehension**: Read the Abstract first → then this section → then Chapter 6 (Discussion: what this paper has accomplished, what it has not, and its representative status) → then Chapter 7 (Conclusion). This pathway captures the core insight and methodological contribution in approximately 20 minutes.
- **For methodological depth**: Read Chapter 3 (The Logical Reversal Chain) and Chapter 4 (Operationalization Protocols) with care. These four sections exhibit the complete operation of the Criterion Dual-Reversal Method and constitute the primary locus of the paper's methodological contribution.
- **For empirical verification**: Proceed directly to Chapter 4 (Operationalization Protocols and Data Sources) and Chapter 5 (Compilation and Presentation of Empirical Materials), to examine the practical feasibility of the operationalization pathway.
- **For concern with grand theory**: Begin with Chapter 1 Introduction, focusing on 1.1 (The Shared Predicament of Grand Theory) and Chapter 6, 6.4 (Representative Status and Open-Interface Character).

Chapter 1 Introduction

1.1 The Shared Predicament of Grand Theory

Grand social theory has always been the highest intellectual form through which human beings attempt to understand their collective existence. Marx's "class consciousness," Weber's "charisma," Durkheim's "social solidarity," Foucault's "dispersion of power"—each of these concepts underpins a grand framework for explaining the collective behavior of human beings. They are rigorously coherent within their theoretical architectures; each can, at the level of concepts, describe a certain systemic state with precision.

Yet on the road toward empirical tractability, grand theory encounters a shared obstacle. How strong is class consciousness? From what point onward does social solidarity begin to loosen? To what degree must power be dispersed before it counts as "dispersed"? These core variables find no direct counterparts in the empirical world. One can argue that a society is experiencing a "crisis of meaning," but one cannot provide a numerical indicator comparable to the GDP growth rate. One can describe a civilization that is losing its "vitality," but one cannot define a unit by which vitality might be independently measured.

This is not an oversight committed by any particular theorist. It is dictated by the very nature of grand theory. What grand theory seeks to capture are often the emergent properties of complex systems—holistic states generated out of innumerable micro-interactions, irreducible to the aggregation of individual units. These states can be clearly defined at the conceptual level, yet their "mode of existence" does not directly correspond to any single observable indicator. "Class consciousness" does not reside inside any one worker's head, nor does it reside in the aggregated data of any survey report. It is a holistic state produced by the cognition, affect, and action of countless individuals coupled together under specific historical conditions. Measuring it directly is impossible in principle—not because the technique is temporarily unavailable, but because it does not belong to the category of things that "can be measured directly."

The same predicament also appears in *Civilization Dynamics*. That theory proposes the concept of Complexity Metabolic Failure (CMF), a term for diagnosing a specific pathological state of civilizational systems: when the incremental information produced by the lateral interactions of subsystems persistently exceeds the sense-making—analysis—response bandwidth of the top-level governance protocols, the system enters a state of functional congestion in which input remains effective but output becomes disordered. The theoretical core of CMF is a dynamic inequality: $C > P \rightarrow$ functional congestion is irreversible. Here, C denotes Interaction Complexity Output, and P denotes Protocol Metabolic Capacity.

This inequality is rigorously coherent within the theory. It is supported by the axiomatic system of *Systems Ontology*, it possesses a clear kernel postulate, and it has well-defined boundaries of validity. But P itself—the information-processing capacity of a system's top-level governance protocols—how is it to be measured? What is the "protocol metabolic capacity" of a society? What is its unit? Above what numerical value is it considered healthy, and below what value is it considered failure?

The CMF theory's answer to this is: do not measure P itself directly. Instead, measure the system-level consequences produced by the insufficiency of P. This strategy was condensed into three draft criteria—Diminishing Returns of Coordination (Criterion A), Response Latency Deadlock (Criterion B), and Antagonism Failure (Criterion C). The three criteria received systematic theoretical construction in the CMF paper, but they have not yet advanced beyond the stage of directional statements. Criterion A states that "the growth rate of macro-coordination costs persistently exceeds the growth rate of total systemic utility"—but what indicator is to serve as a proxy for coordination costs? Criterion B states that "the typical response latency to critical interaction events persistently exceeds the crisis half-life of such events"—but how is the half-life to be calibrated, and how is the response cycle to be demarcated? Criterion C states that "the explanatory coverage of the meta-narrative exhibits irreversible attenuation"—but what is to serve as the measure of the coverage of a meta-narrative?

If these questions remain unanswered, CMF will remain stuck in the traditional form of grand theory: conceptually self-consistent, yet with criteria that are not operationalizable. It can use the scalpel of theory to dissect the pathologies of the world, but it cannot prove to the skeptic that what it is dissecting is indeed a lesion, and not a phantom of its own imagining.

1.2 The Task of This Paper: A Proof of Concept

The task of this paper is not to prove the correctness of CMF theory. The correctness of a theory requires successive rounds of independent testing, cross-case comparison, and cross-criterion coupling before it can be progressively confirmed. This is not something a single paper can accomplish.

Nor is the task of this paper to conduct an empirical test of the CMF criteria. The empirical testing of the three criteria requires that the failure thresholds first be calibrated—where lies the "point of diminishing returns" for Criterion A? How is the "intolerable latency threshold" for Criterion B to be demarcated? At what threshold value does the "loss of coverage" of Criterion C constitute "failure"? The calibration of these thresholds is itself an independent research project, one that requires large volumes of data and repeated rounds of calibration. Before the thresholds are calibrated, the question "Is failure currently occurring?" is one that cannot be empirically validated.

The task of this paper is more foundational than either of the above. The question this paper seeks to answer is this: **Can the three criteria of CMF be transformed from directional statements into operationalizable indicators?** Does this transformation pathway exist? Can it be traversed by an independent researcher under resource constraints?

This is not a theoretical question. It is an engineering question. It does not ask "Are the criteria correctly designed?"—answering that requires empirical testing. It asks "Can this theoretical apparatus be made to touch the ground?"—answering that requires actually performing operationalizations, locating data, and verifying that the data are indeed usable.

Thus, this paper is positioned as a **proof of concept**. As with proof-of-concept experiments in physics or engineering, the aim is not to show that "this thing works under all conditions," but to show that "this thing can be realized under specified controlled conditions." If it cannot be realized, then all subsequent empirical testing hangs in the air. If it can be realized, subsequent researchers will have an operationalization pathway to draw upon.

This paper selects the construction industry as a single-sector proving ground, and sets 2000–2020 as the unified analytical time window. The choice of the construction industry is not because it is the only domain in which operationalization is possible—on the contrary, other regulation-intensive sectors (such as food safety and environmental protection) possess comparable operationalization potential. The construction industry is chosen because it offers the optimal combination of data accessibility for the three criteria: its safety-standard system possesses a complete public record on the National Standardization Administration platform; its typical safety accidents possess clearly traceable temporal anchors in official notifications; and its workforce demographic structure and income data are available in continuous time series in the *Migrant Worker Monitoring Survey Reports*. For an independent researcher, the construction industry is the operationalization site that is most feasible under present conditions and least dependent on high-access data privileges.

Most critically, the author has spent an extended period rooted in the front lines of construction production, and possesses first-hand practical experience and a depth of domain expertise in this industry.

1.3 The Methodological Core: Logical Reversal

The core operation this paper deploys for transforming the criteria is logical reversal.

When a grand-theoretical variable cannot itself be measured directly, one refrains from measuring the variable itself, and instead measures the system-level consequences produced by the absence of that variable. The "meta-cognitive capacity" of a society may be impossible to measure directly, but when it undergoes sustained attenuation, it leaves observable traces at the meaning-layer of the system—for example, the declining intergenerational explanatory coverage of the meta-narrative. Measuring such a trace is equivalent to indirectly calibrating the "degree of absence of meta-cognitive capacity." And the degree of absence, inversely, carries information about the degree of health.

Logical reversal is not an independently-standing operation. It is the core mechanism of the Criterion Dual-Reversal Method. This method consists of two steps: Step 1, deriving from the deep dynamic inequality of the pathological state the criterion signals that necessarily manifest at the observable level; Step 2, reversing the direction of the pathological inequality to obtain the criterion set for the healthy state. What this paper executes, in its operationalization of the three CMF criteria, is precisely these two steps—the reversal logic of each step will be fully exhibited in Chapter 3.

The relationship between the Criterion Dual-Reversal Method and the Boundary Anchoring Method is one of generation and deployment. The former generates criteria—it derives observable signals from abstract pathological concepts. The latter deploys criteria—once criteria have been independently calibrated, it takes them as anchors and calculates the deviation degree of a system under test relative to those anchors. Together, they constitute the complete methodological pathway from grand theory to empirical tractability.

Within the paper sequence to which the present paper belongs, the sorting capacity of the Boundary Anchoring Method has already received independent verification through the "involution verification"—the deviation degrees of three agricultural systems aligned in direction with independent scholarly baselines. Yet that verification rests on an implicit premise: the criteria had already been independently calibrated by Huang Zongzhi. What the present paper seeks to supply is precisely this premise—to demonstrate that the operationalization and calibration of criteria can itself be independently accomplished.

The relationship between the two papers is therefore complementary. The involution verification demonstrated: when criteria are clearly measurable, the Boundary Anchoring Method can effectively locate system states. This paper demonstrates: when criteria remain as directional statements, the Criterion Dual-Reversal Method can transform them into clearly measurable operational indicators. Taken together, they constitute the complete methodological chain from grand theory to empirical tractability.

Chapter 2 Methodological Background

2.1 Two Methodological Instruments and Their Relationship

The methodological foundation of this paper rests on two instruments.

The Criterion Dual-Reversal Method

The Criterion Dual-Reversal Method is responsible for generating criteria. Its core operation consists of two steps.

Step 1: Deriving, from the deep dynamic inequality of the pathological state, the criterion signals that necessarily manifest at the observable level. If a system is sliding toward metabolic failure, its Protocol Metabolic Capacity P is being persistently exceeded by its Interaction Complexity Output C—and this abstract pathology inevitably leaves observable traces across different layers of the system: at the efficiency layer, the input-output ratio of coordination deteriorates; at the temporal layer, response latency outstrips the speed at which crises diffuse; at the meaning layer, the cohesive force of the meta-narrative degrades from a default consensus into something requiring external compulsion to maintain. These traces are the criteria.

Step 2: Reversing the direction of the pathological inequality to obtain the criterion set for the healthy state. If the mark of failure is "the growth rate of coordination costs exceeds the growth rate of utility," then the mark of health is "the growth rate of coordination costs persistently falls below the growth rate of utility."

The Boundary Anchoring Method

The Boundary Anchoring Method is responsible for deploying criteria. Its core operation is as follows: once criteria have been independently calibrated, they are taken as anchors, and the deviation degree of a system under test relative to those anchors is calculated. The calculation of the deviation degree is itself an independent output of the Boundary Anchoring Method—it does not require additional comparison with independent baselines to count as "complete." Whether that output is consistent with independent scholarly baselines belongs to the verification agenda of the method's self-validation phase and does not constitute a defining step of the method itself.

The division of labor between the two is thus clear. The question answered by the Criterion Dual-Reversal Method is: "Where do criteria come from? On what grounds do we believe that a given observable signal carries information about the health status of the system?" The question answered by the Boundary Anchoring Method is: "Once we have such a criterion, how do we use it to quantitatively locate the current state of a specific system?" The former generates criteria; the latter deploys them. Together, they constitute the complete methodological pathway from grand theory to empirical tractability.

2.2 The Boundary Anchoring Method Has Already Been Independently Verified

Prior to this paper, the validity of the Boundary Anchoring Method has already been tested through an independent verification.

That verification took agricultural involution as its case, adopted the unit labor-day remuneration of Jiangnan rice farming as the left-end anchor—as given in the original work of Huang Zongzhi—used the ratio-method formula as its computational instrument, and completed a differentiated calculation of the involution deviation degree for three systems: the Jiangnan cotton-textile sector at 19.5%, the North China poor-peasant family farm at 42%, and the Tokugawa Kinai region trending upward, away from the limit. The computational results themselves constituted the independent output of that study. The subsequent comparison with independent scholarly baselines in agricultural historiography—according to which Jiangnan suffered the deepest involution, North China was less severe or of a different type, and Kinai represented non-involutionary intensive development—was an external verification of that output; the ordinal ranking of the three systems aligned in direction with the independent baselines.

The conclusion of that study was: when criteria are clearly calibrated, the Boundary Anchoring Method can effectively calculate deviation degrees and produce a quantitative location of system states.

Yet that study rested on an implicit premise: the left-end anchor—that is, the decay limit of involution—had already been independently calibrated by Huang Zongzhi. Huang not only supplied the original numerical values for rice farming as the "non-involution benchmark," but also directly computed the labor-remuneration ratio of the cotton-textile sideline relative to that benchmark. What the involution verification demonstrated was the computational logic of the Boundary Anchoring Method—its capacity, once fed calibrated criteria as input, to effectively produce a location result. What it did not address was a logically prior question: **How do the criteria themselves advance from directional statements to operational calibration?**

2.3 The Lacuna This Paper Seeks to Fill

The three criteria of Complexity Metabolic Failure remain, in the CMF paper, at the stage of directional statements. They indicate the direction in which to observe—"the growth rate of coordination costs exceeds the growth rate of utility," "response latency exceeds the speed of crisis diffusion," "the explanatory coverage of the meta-narrative attenuates"—but they supply no concretely executable measurement protocol. What is the proxy variable for Criterion A? From what starting point and to what endpoint is the response latency of Criterion B to be calculated? What signal is to be used to track the attenuation of explanatory coverage in Criterion C? If these questions are not resolved, the Boundary Anchoring Method cannot take over—without criterion input, the deviation-degree calculation cannot be initiated.

The task of this paper is precisely to accomplish, using the Criterion Dual-Reversal Method, the first operational calibration of these three criteria. This paper is the second in the verification sequence of the Boundary Anchoring Method. The involution verification demonstrated: when criteria are clearly measurable, the Boundary Anchoring Method can effectively locate system states. This paper demonstrates: when criteria remain as directional statements, the Criterion Dual-Reversal Method can transform them into clearly measurable operational indicators. Taken together, they constitute the complete methodological chain from grand theory to empirical tractability.

2.4 The Methodological Execution Pathway of This Paper

This paper executes the complete operational procedure of the Criterion Dual-Reversal Method in three steps.

Step 1: Logical reversal. For each criterion, follow the four-step logical chain of "original predicament → key reversal → preconditions → operationalization result." "Original predicament" explains why the core variable of that criterion cannot be measured directly. "Key reversal" explains where the thinking pivots, displacing "measuring the variable itself" with "measuring the consequences of the variable's absence." "Preconditions" openly declares the presuppositions on which the validity of the reversal rests. "Operationalization result" gives the observable signal ultimately locked in after the reversal. This step receives a complete exhibition in Chapter 3.

Step 2: Designing the operationalization protocol. After the logical reversal has established "what to measure," for each criterion, select a proxy variable, confirm the data sources, and state the analytical method. The selection of proxy variables follows a proximity principle: among multiple feasible candidate indicators, priority is given to the one whose data are publicly available, whose time series is complete, and which can be independently replicated by other researchers. This step is completed in Chapter 4.

Step 3: Obtaining and organizing empirical materials. Taking the construction industry as the single-sector proving ground and 2000–2020 as the unified temporal window, obtain empirical data for each criterion according to the operationalization protocol. This paper does not claim that these materials "prove" that failure is currently occurring—the thresholds have not yet been calibrated, and this judgment cannot be made. This paper demonstrates only that: according to the present protocol, the data are obtainable; and the trends exhibited by the obtained data are broadly consistent with the theoretically-expected direction of each criterion. This step is completed in Chapter 5.

After the entire operation is complete, this paper does not perform a deviation-degree calculation. The endpoint of this paper lies at the point where criterion operationalization and calibration is completed. Subsequent researchers may take the criteria calibrated in this paper and directly feed them into the

Boundary Anchoring Method to calculate deviation degrees—the calculation of deviation degrees is itself an independent output of that method, a quantitative operationalization of the system parameters to which the criteria point. Whether that output is consistent with independent scholarly baselines belongs to the verification agenda of the method's self-validation phase and does not constitute the task of this paper.

2.5 The Research Boundaries of This Paper

The research boundaries of this paper must be declared here in advance.

First, this paper does not calibrate failure thresholds. The "point of diminishing returns" for Criterion A, the "intolerable latency threshold" for Criterion B, the "coverage-loss threshold" for Criterion C—the calibration of these thresholds requires more accumulated cases and iterative parameter adjustment, and is not something a single proof-of-concept paper can accomplish.

Second, this paper does not perform cross-criterion coupling and integration. The three criteria each independently undergo operationalization; after completion, they are not combined into a weighted composite or synthetic judgment.

Third, this paper does not generalize its conclusions to industries beyond the construction sector.

Fourth, this paper does not perform a deviation-degree calculation.

The function of these declarations is to clearly demarcate the boundaries of a proof of concept. The fatal risk for a proof-of-concept paper is that the reader mistakes "the concept has been proven" for "the theory has been confirmed." The former requires only demonstrating the existence of a pathway; the latter requires demonstrating the correctness of a conclusion. What this paper accomplishes is the former.

Chapter 3 The Logical Reversal Chain of Criterion Operationalization

This chapter constitutes the methodological core of the entire paper. Each of the three criteria follows the four-step logical chain of "original predicament → key reversal → preconditions → operationalization result."

3.1 Criterion A: From "Total Coordination Cost" to "Rule-Revision Frequency"

3.1.1 Original Predicament

The core variable of Criterion A is "macro-coordination cost." In CMF theory, this denotes the total cost a system expends to maintain its overall coordination function—the sum total of rule-making and rule-revision, compliance monitoring, conflict mediation and dispute resolution, and cross-departmental coordination. The original statement of the CMF paper is: when the growth rate of this cost persistently exceeds the growth rate of total systemic utility, Criterion A turns positive.

The predicament is that no ready-made statistical indicator for "total coordination cost" exists in the social sciences. Accounting ledgers contain no entry for "coordination costs"; the national economic accounting system includes no line item for "rule-maintenance expenditures." To measure the total coordination cost of a societal system directly, one would need to simultaneously track cost occurrences across multiple sites—legislative bodies, regulatory agencies, court systems, industry self-regulatory organizations, corporate compliance departments—and these costs are dispersed across incommensurable dimensions such as monetary expenditure, labor input, and time consumption, making aggregation into a single numerical value impossible.

This is not a problem of data-access difficulty. It is a categorical feature of the concept "coordination cost" itself—it is a functional burden diffused across every node of the system, not a variable that can be collected centrally.

3.1.2 Key Reversal

Shift from "measuring the cost itself" to "measuring the driver of the cost."

Why do coordination costs rise? In complex social systems, the principal carrier of coordination costs is rules—laws, regulations, industry standards, and procedural codes. The function of rules is to constrain the behavioral boundaries of subsystems, rendering their interactions predictable. When reality repeatedly breaches the boundaries of old rules—new technologies emerge, new business models appear, new safety risks are exposed—the system is compelled to revise its rules repeatedly in order to re-accommodate reality. Each revision represents an instance of coordination activity. A surge in revision frequency signals that "new events requiring coordination" are emerging in dense succession.

Thus, rather than computing total coordination costs directly, one tracks the frequency and magnitude of rule revisions. The long-term trend of revision frequency carries valid information about whether the coordination burden is intensifying.

3.1.3 Preconditions

For this reversal to be valid, two preconditions must be satisfied.

First, rule revision is not a matter of jurists performing meticulous refinement in their chambers. Large-scale, high-frequency revision is the result of reality having already breached the boundaries of old rules. When a given domain, over a window spanning more than a decade, sees revision frequency leap from "one item every few years" to "several items or even more than ten items every year," this cannot be attributed solely to "legislators suddenly becoming more diligent." A more plausible interpretation is that the rate of real-world change in that domain has outstripped the covering capacity of existing rules, and the rule system is being repeatedly buffeted by reality.

Second, the rules whose revisions are counted must possess real binding force, not be merely hortatory "soft law." Non-binding advisory documents, regardless of how many are issued, do not constitute valid indicators of coordination cost. The mandatory national standards and industry standards selected in this paper all contain compulsory provisions, and violators face administrative penalties or legal liability. Once promulgated or revised, such standards mandatorily alter the operational procedures and cost structures of the enterprises concerned. Their revision events thus carry genuine information about coordination costs.

3.1.4 Operationalization Result

The core proxy variable for Criterion A is the annual revision count of mandatory national standards and industry standards related to construction safety. The analytical method is: extract the number of new standards issued and existing standards revised in each year from 2000 to 2020, and observe the long-term trend of revision frequency—whether it is steady, sporadic, or persistently accelerating. If the revision frequency exhibits a marked upsurge from "stable low frequency" to "persistent high frequency" over the time series, then Criterion A receives preliminary empirical support.

3.2 Criterion B: From "Half-Life" to "First Victim Node"

3.2.1 Original Predicament

The core variable of Criterion B is the "crisis half-life"—the length of time it would take, were a hazardous event left unaddressed, for its negative consequences to spontaneously diffuse throughout the entire system. The original statement of the CMF paper is: when the system's typical response latency for critical interaction events persistently exceeds the crisis half-life of such events, Criterion B turns positive.

The concept of "half-life" is borrowed from nuclear physics and possesses no natural counterpart in the social sciences. For a societal hazard—say, a particular violation pattern in a high-risk industry, or the emission of a particular pollutant—what exactly does "diffusion throughout the entire system" mean? How broad a scope must the diffusion reach, how large a population must it affect, and what degree of irreversibility must it attain before it counts as "having run its course"? These questions have no consensual answers. Half-life cannot be observed directly—it must be estimated.

3.2.2 Key Reversal

Shift from "estimating the half-life" to "defining the failure threshold by reference to the first victim case."

Abandon the abstract definition of "diffusion throughout the system" and instead lock onto an indisputable temporal node: the date on which the first publicly-reported paradigmatic victim case appears. Prior to this date, the absence of a regulatory response may be attributed to the fact that "the response is still within a reasonable window"—because the hazard has not yet been publicly demonstrated to be real, concrete, and carrying substantive consequences. After this date, if the regulatory response still has not materialized, one may judge that "response failure" has occurred—because the system failed to complete timely self-correction after the hazard had been empirically confirmed as an open event involving specific victims.

Rather than attempting to compute an abstract diffusion duration, one simply compares two concrete dates: the date on which the hazard was first authoritatively and definitively designated by an official body, and the date on which a targeted regulatory instrument formally entered into force. The actual response latency is that interval itself. The first publicly-reported paradigmatic victim case serves as the critical threshold for judging whether the response was tolerable—if the regulatory instrument was completed before the arrival of the critical threshold, the system was operating within a normal response cycle; if the instrument arrived only after the first victim, the system had already entered response latency deadlock.

3.2.3 Preconditions

For this reversal to be valid, two preconditions must be satisfied.

First, zero response latency is unrealistic. A societal system cannot be expected to legislate preemptively before the causal mechanism of a hazard has even been scientifically identified—this would demand infinite foresight on the part of decision-makers and is physically impossible. Hence, the absence of a regulatory response before the hazard mechanism is scientifically established cannot be used to judge Criterion B positive. The starting point of the response window should be set at the moment the authoritative body first definitively designates the existence of the hazard, not at the moment the hazard first physically appeared.

Second, the first publicly-reported paradigmatic victim case is a node that can be clearly demarcated using official notifications. It is not equivalent to "the first actual victim event that occurred"—the first actual victim may have gone unrecorded. It denotes the first typical accident that was formally notified by a national authoritative body and written into an annual safety-warning case compendium. Once this node appears, it means: the hazard is no longer merely a statistical probability in a scientific report, but an open, socialized event that the public can recognize as real. If effective regulatory response remains absent beyond this critical threshold, the failure is no longer attributable to "cognitive limitations" but to "response-mechanism failure."

3.2.4 Operationalization Result

Take two hazard types within the domain of construction safety that have been independently confirmed by research—falls from height and foundation pit collapses—as parallel cases, and demarcate three key dates for each: the issuance date of the official document in which the authoritative body first definitively designated the hazard; the official notification date of the first publicly-reported paradigmatic victim accident; and the formal entry-into-force date of the targeted standard or revised provision. The response latency is the interval between the entry-into-force date of the targeted standard and the authoritative designation date. Compare the response latency ordinally with the date of the first paradigmatic victim case: if the regulatory response was completed before the first victim, the system was within a normal response interval; if it was completed only after the first victim, Criterion B receives preliminary empirical support.

3.3 Criterion C: From "Meta-Narrative Coverage" to "Reverse Maintenance Signals"

3.3.1 Original Predicament

The core variable of Criterion C is the "explanatory coverage of the meta-narrative." The CMF paper defines the meta-narrative as the highest-level cognitive framework shared by a civilization or large-scale social system—it explains "who we are, why we are here, and where we are heading." When the explanatory coverage of this framework exhibits a monotonically decreasing trend over intergenerational observation cycles, Criterion C turns positive.

The meta-narrative manifests at the societal level in multiple forms: the basic unit of society shifting from the traditional clan family toward individualization; marital and familial norms undergoing continuous renovation; systemic ruptures emerging between the occupational choices and life-meaning narratives of successive generations. These are all concrete manifestations of meta-narrative iteration. Yet they are all qualitative descriptions—"explanatory coverage" itself is an emergent property, and cannot be observed directly in the way that, say, literacy rates can.

3.3.2 Key Reversal

Shift from "measuring meta-narrative coverage itself" to "measuring the reverse signals that appear when meta-narrative coverage has been lost."

A healthy meta-narrative does not need to be repeatedly emphasized. When a narrative truly permeates the capillaries of social life, it is the cognitive framework that the public defaults to—like water to a fish, or air to a human being, those immersed in it do not consciously notice its presence. Only when the existing narrative system begins to fail to match reality, and contradictions keep surfacing, do people become aware of the limitations of the old rules and concepts. At this point, the institutions that maintain the old narrative develop a "maintenance impulse"—they reinforce the narrative through official discourse, policy advocacy, and the education system. The intensity, frequency, and rhetorical form of this maintenance are observable signals.

Thus, "the meta-narrative is being repeatedly, pointedly, and even coercively maintained"—this is the reverse signal of declining coverage. This signal does not directly measure the absolute value of coverage; what it measures is the surge in the need for maintenance. And a surge in the need for maintenance carries valid information about the fact that coverage is being lost.

3.3.3 Preconditions

For this reversal to be valid, two preconditions must be satisfied.

First, there exists an intrinsic adaptive relationship between a governance apparatus and the social structure and dominant meta-narrative in which it is embedded. It is not a passive recipient of a narrative; rather, it actively maintains the kind of narrative most conducive to its own stability and to the functioning of the system. "Hard work brings prosperity," "Safety first in production"—these narratives are advocated not because they are somehow more "correct" in a vacuum, but because the modes of social behavior they underpin effectively sustain the stable operation of the system. When these narratives begin to loosen at the societal level, the response of the governance apparatus is not neutral—it develops a structural maintenance impulse.

Second, the intensity and rhetorical form of this maintenance impulse are observable and documentable. Healthy advocacy is unobtrusive—it permeates textbooks, news reporting, and public rituals without needing to be labeled "propaganda." Defensive reinforcement, by contrast, possesses markedly different rhetorical features: the frequent deployment of compulsory vocabulary; the repeated invocation of attribution frames in which the narrative is allegedly being challenged by "erroneous currents"; and the linking of narrative compliance with institutional consequences. This shift in rhetorical form is itself a signal capturable through discourse analysis.

3.3.4 Operationalization Result

Two complementary information streams serve as the core proxy variables for Criterion C. The first stream is macro-level data: track changes in the intergenerational structure of construction-industry migrant workers—whether the proportion of older workers is systematically rising and young labor is undergoing net outflow; and track changes in the ratio of construction-industry wages to the average price of commercial housing—whether the 兑现 capacity of the traditional "hard work brings prosperity" narrative at the purchasing-power level is attenuating. The second stream is discourse analysis: track the rhetorical-form shifts of the core narrative of "safety in production" in official discourse—from "Safety first, prevention foremost" circa 2000, to "Life supreme, red-line consciousness" post-2010, and assess whether a shift from advocacy-oriented to defense-oriented discourse has occurred. If the objective data stream reflects generational rupture and purchasing-power attenuation, and the discourse-analysis stream reflects a persistent escalation of defensive maintenance, the cross-verification of the two provides preliminary empirical support for Criterion C.

Chapter 4 Operationalization Protocols and Data Sources

Having established "what to measure" through the logical reversals of Chapter 3, this chapter selects proxy variables, confirms data sources, and states analytical methods for each criterion.

4.1 Operationalization Protocol for Criterion A

4.1.1 Proxy Variable

The annual revision count of mandatory national standards and industry standards in the domain of construction safety, together with the case-specific revision timelines of representative standards.

4.1.2 Data Sources

The National Standard Information Service Platform and the official website of the Ministry of Housing and Urban-Rural Development. The former provides the promulgation and revision dates of standards; the latter supplies the annual *Work Plans for the Formulation and Revision of Engineering Construction Standards* and the official promulgation notices for individual standards. The complete revision histories of typical standards—for example, JGJ 80 *Technical Code for Safety of Work at Heights in Building Construction* and JGJ 59 *Standard for Safety Inspection of Building Construction*—can be obtained by searching the standard number on the standards platform.

4.1.3 Analytical Method

First, extract the annual revision count of construction-safety-related standards for the period 2000–2020. Taking two- to three-year intervals, compile the number of standards newly issued, revised, and repealed in each year, and observe the long-term trend of revision frequency. If the revision frequency exhibits a marked upsurge from "stable low frequency" to "persistent high frequency," then Criterion A receives preliminary empirical support.

Second, select representative standards for case-specific analysis of revision timelines. JGJ 80-91 was implemented in 1992, and its successor JGJ 80-2016 was implemented in 2016, an interval of 24 years. JGJ 59-99 was succeeded by JGJ 59-2011 after an interval of 12 years. Case-specific analysis, in conjunction with the annual revision-count trend, constitutes a dual body of evidence combining macro-level trends with micro-level instances.

4.2 Operationalization Protocol for Criterion B

4.2.1 Proxy Variable

Two hazard types within the domain of construction safety that have been independently confirmed by research—falls from height and foundation pit collapses—are selected as parallel cases. For each, three key dates are demarcated: the issuance date of the official document in which the authoritative body first definitively designated the hazard; the official notification date of the first publicly-reported paradigmatic victim accident; and the formal entry-into-force date of the targeted standard or revised provision.

4.2.2 Data Sources

The authoritative hazard-designation document is Document JZ [2003] No. 82, "Notice on Issuing the *Several Provisions on the Prevention of Falls-from-Height Accidents in Building Construction* and the *Several Provisions on the Prevention of Collapse Accidents in Building Construction*," issued by the former Ministry of Construction on April 17, 2003. This document is archived on the official website of the Ministry of Housing and Urban-Rural Development as well as on the websites of various local housing and construction authorities. The targeted standard for falls from height, JGJ 80-2016, was approved by Ministry of Housing and Urban-Rural Development Proclamation No. 1205; the targeted standard for foundation pit collapses, GB 50202-2018, was approved by Ministry Proclamation No. 23 of 2018. Official notifications of typical accidents are retrievable from the official websites of the Ministry of Housing and Urban-Rural Development and the Ministry of Emergency Management.

4.2.3 Analytical Method

Compute the response latency for each of the two hazard types. The response latency is the interval between the formal entry-into-force date of the targeted standard and the authoritative designation date. Compare the response latency ordinally with the date of the first paradigmatic victim case: if the regulatory instrument was issued only after the first paradigmatic victim had already appeared, this indicates that the system failed to complete timely regulatory response even after the hazard had been openly confirmed as a concrete victim-involving event—Criterion B receives preliminary empirical support. If two different hazard types exhibit response cycles of a similar order of magnitude, the weight of case-specific idiosyncrasy is reduced, and the weight of shared regularity is strengthened.

4.3 Operationalization Protocol for Criterion C

4.3.1 Proxy Variable

Two complementary information streams are adopted. The first is a macro-level data stream: changes in the intergenerational structure of migrant workers in the construction industry, and changes in the ratio of construction-industry wages to the average price of commercial housing. The second is a discourse-analysis stream: shifts in the rhetorical form of the core narrative of "safety in production" in official discourse.

4.3.2 Data Sources

The National Bureau of Statistics has issued the *Migrant Worker Monitoring Survey Report* annually since 2008. This report supplies data on the age structure, sectoral distribution, and income of migrant workers in the construction industry; detailed indicators for selected years extend back before 2010. The *China Labor Statistical Yearbook* supplies the continuous time series of the average wage of on-post workers in the construction industry. The annual data of the National Bureau of Statistics supply the nationwide average sales price of commercial housing. Materials for official discourse analysis are drawn from safety-accident notifications and annual work reports issued on the official website of the Ministry of Housing and Urban-Rural Development, as well as from "safety in production" thematic reports carried on official news platforms.

4.3.3 Analytical Method

Dimension 1: Quantitative analysis of generational rupture. Extract the time series of the proportion of construction-industry migrant workers aged 50 and above for the period 2000–2020. If this proportion exhibits a persistent upward trend, and the proportion of young workers simultaneously exhibits a persistent decline, this indicates that the attractiveness of the industry is attenuating and that a generational exodus of labor is underway. This is the behavioral signal that the narrative of "hard work brings prosperity" is losing its explanatory power within this industry.

Dimension 2: Quantitative analysis of the purchasing-power ratio. Compute the ratio of the median wage of construction-industry workers to the median price per square meter of commercial housing over the same period. If this ratio declines persistently from 2000 to 2020, this indicates that, even as nominal wages rise, real purchasing power is undergoing relative attenuation. This is the attenuation of the explanatory power of the "hard work brings prosperity" narrative at the level of material fulfillment of its promises.

Dimension 3: Qualitative analysis of the defensive turn in official discourse. Track shifts in the rhetorical form of the core narrative of "safety in production" in official discourse. The dominant discourse in the years around 2000 was "Safety first, prevention foremost" and "Strictly prohibit operations in violation of regulations"—a command-style, penalty-oriented first-generation discourse. The dominant discourse after 2010 shifted toward "Life supreme," "Red-line consciousness," and "Zero tolerance"—a defense-reinforcement, moralization-escalation third-generation discourse. The escalation of discourse from "advocacy" to "defense" constitutes auxiliary evidence of the declining coverage of the meta-narrative.

Cross-verification of the three dimensions. If the generational data reflect a systemic exit of young labor, the purchasing-power data reflect a persistent attenuation of the material 兑现 capacity of "hard work brings prosperity," and the discourse analysis reflects a shift in official maintenance of the "safety in production" narrative from advocacy to defensive reinforcement—with all three information streams converging in direction—then Criterion C receives preliminary empirical support. The causal direction here is: the attenuation of narrative explanatory power is not the result of young people no longer believing that hard work can bring prosperity. Rather, the conditions of social reality have changed, such that the narrative, which originally depended upon those conditions for its validity, can no longer deliver on its promise. It is not that ideas changed first; it is that reality changed first. The judgment that young labor enacts through exit is not a rejection of the narrative as such, but a decision not to enter after assessing reality. The defensive turn in official discourse is a reaction to this social reality, not its cause.

4.4 Attribute Declaration Concerning the Empirical Data

All data listed in this chapter derive from publicly available government documents and official statistical platforms. The coverage and granularity of these data are subject to the resource conditions of an independent researcher. This paper cannot, as a large-scale research team might, apply for access to non-public administrative microdata, nor can it purchase commercial datasets. Yet the principle by which this paper selects proxy variables—就近 selecting indicators whose data are publicly available, whose time series are complete, and which can be independently replicated by other researchers—ensures that the operationalization protocol is fully executable within the resource boundaries of an independent researcher. The task of Chapter 4 is not to exhaust all available proxy variables, but to identify, for each criterion, a first usable operationalization pathway under present resource constraints.

4.4.1 The Researcher's Professional Credentials and Industry-Cognition Base

In selecting the construction industry as the operationalization proving ground, the researcher possesses a capacity for professional judgment that extends beyond the reading of the literature. The author spent an extended period on the front lines of production and possesses first-hand practical experience and domain knowledge of how construction safety management actually operates. Furthermore, the author holds a nationally-certified professional qualification. This qualification is not merely a legal authorization to practice; it also signifies that the author has undergone systematic professional training and rigorous competency examinations—and hence possesses a cognitive-judgmental grasp, difficult for those outside the industry to acquire, of the regulatory framework of the construction-safety domain, the periodicity of standard revision, the typical patterns in which accidents occur, and the structural characteristics of the workforce.

The selection of the construction industry as the operationalization proving ground is, at the level of research design, a comprehensive judgment based on data availability and sectoral suitability. At the level of research capacity, it rests on the author's deep experiential knowledge of this industry. Matters that require experiential judgment—whether a shift in the tempo of standard revision reflects routine updating or a genuine intensification of sectoral pressures; whether the accident cases selected possess typicality rather than being merely sporadic; whether the data on generational exit genuinely point to structural decline or merely to short-term fluctuations—are all calibrated by the researcher's first-hand industry cognition. Such calibration does not substitute for logical inference from the data, but it ensures that the interpretation of the data does not depart from the real operative logic of the industry.

Chapter 5 Compilation and Presentation of Empirical Materials

This chapter presents the results of the empirical material acquisition following the operationalization protocols set out in Chapter 4. This chapter does not claim that these materials "prove" that failure is currently occurring—the thresholds have not yet been calibrated, and this judgment cannot be made. This chapter demonstrates only that: according to the present protocols, the empirical materials are obtainable; and the trends exhibited by the obtained materials are broadly consistent with the theoretically-expected direction of each criterion.

5.1 Criterion A: The Upward Surge in Standard Revision Frequency

5.1.1 Material Acquisition

The standards system in the construction-safety domain is composed of mandatory national standards (the GB 50XXX series) and industry standards (the JGJ series). Both categories contain compulsory provisions, and their promulgation and revision records are publicly available in full on the official website of the Ministry of Housing and Urban-Rural Development and on the National Standard Information Service Platform.

At the level of macro-level trends, the Ministry of Housing and Urban-Rural Development issues an annual *Work Plan for the Formulation and Revision of Engineering Construction Standards*, listing the standards to be newly compiled, revised, or repealed in the given year. Taking the year 2020 as an example, Ministry Document JBH [2020] No. 9 states explicitly: "In order to implement the reform requirements for the engineering construction standards system, promote the high-quality development of engineering construction, safeguard the quality and safety of engineering works, improve people's livelihoods, and protect the ecological environment, our Ministry has organized the formulation of the *2020 Work Plan for the Formulation and Revision of Engineering Construction Codes and Standards*." Documents of this type, issued annually, constitute a continuous time series for tracking the frequency of standard revisions.

At the level of individual cases, two standards foundational to the industry provide a complete trajectory of revision. JGJ 80 *Technical Code for Safety of Work at Heights in Building Construction* was first promulgated in 1992 as JGJ 80-91 and was revised in 2016 as JGJ 80-2016, approved for implementation by Ministry of Housing and Urban-Rural Development Proclamation No. 1205—an interval of 24 years between the old and new versions. JGJ 59 *Standard for Safety Inspection of Building Construction* was promulgated in 1999 as JGJ 59-99 and revised in 2011 as JGJ 59-2011, approved for implementation by Ministry Proclamation No. 1204—an interval of 12 years.

5.1.2 Trend Observations

Within the unified temporal window of 2000–2020, the revision frequency of construction-safety standards exhibits a pronounced pattern of "initially steady, subsequently surging." During the period 2000–2010, the annual revision count of safety-related standards lay approximately within the range of 5–10 items, and the revision tempo was relatively balanced. After 2011, the annual revision count began to increase markedly, and after 2015 entered a period of high-density revision.

More significant is the structural shift in the driving force behind the revisions. Revisions prior to 2010 were dominated by routine refinement of the standards system. After 2010, an increasing proportion of revisions came to correspond directly to specific categories of safety accidents or newly-exposed systemic risks—"problem-driven revision" gradually became dominant. The upward surge in revision frequency signals that new events requiring re-accommodation by the rules are emerging in dense succession.

5.2 Criterion B: Cross-Hazard-Type Measurement of Response Latency

5.2.1 Material Acquisition

On April 17, 2003, the former Ministry of Construction issued Document JZ [2003] No. 82, printing and distributing the *Several Provisions on the Prevention of Falls-from-Height Accidents in Building Construction* and the *Several Provisions on the Prevention of Collapse Accidents in Building Construction*. The documents explicitly designated falls from height and foundation pit collapses as major hazard types for prevention and control in the construction industry. This is the common starting point of authoritative designation for both hazard types.

The targeted response for falls from height is marked by the issuance of JGJ 80-2016. Ministry of Housing and Urban-Rural Development Proclamation No. 1205 (issued July 9, 2016) approved this code as an industry standard, to enter into force on December 1, 2016, replacing JGJ 80-91, which had been in effect since 1992. From the authoritative designation in 2003 to the implementation of the targeted standard in 2016, the response latency is approximately 13 years. During this interval, the November 11, 2004, catastrophic fall-from-height accident in Dongguan was nationally notified by the Ministry of Housing and Urban-Rural Development as **a nationally-designated model case subject to special supervision**, becoming a benchmark case written into the annual safety-warning compendium—the first publicly-reported paradigmatic victim case occurred merely one year after the authoritative designation.

The targeted response for foundation pit collapses is marked by the issuance of GB 50202-2018. Ministry of Housing and Urban-Rural Development Proclamation No. 23 of 2018 approved this code as a national standard, to enter into force on October 1, 2018, replacing the 2002 version. From the authoritative designation in 2003 to the implementation of the targeted standard in 2018, the response latency is approximately 15 years.

5.2.2 Trend Observations

The response latencies for the two hazard types are approximately 13 and 15 years, respectively. Both are of the same order of magnitude—in each case, exceeding a decade. This cross-hazard-type similarity reduces the explanatory weight of "case-specific idiosyncrasy" and strengthens the plausibility of the interpretation that the response mechanism suffers from a structural lag.

The span of the response latency itself far exceeds the life cycle of any single construction project. During the 13 or 15 years it took for a standard to progress from being recognized as needing revision to actually completing revision, the industry had already undergone several complete cycles of technological iteration, and tens of millions of workers had been exposed to hazards that had already been definitively designated by official bodies. The initiation of regulatory repair occurred not at the moment the hazard was designated, nor at the moment the first member of the public became a victim, but only after decades of accumulated sectoral pressures and incremental information had built up.

5.3 Criterion C: Generational Rupture, Purchasing-Power Attenuation, and Discursive Shift in

the Construction Industry

5.3.1 Material Acquisition

The National Bureau of Statistics has issued the *Migrant Worker Monitoring Survey Report* annually since 2008. This report lists the construction industry as one of six principal monitored sectors and supplies data on the age structure, sectoral distribution, and income of workers in the sector. The annual data of the National Bureau of Statistics and the *China Labor Statistical Yearbook* supply, respectively, the continuous time series of the average sales price of commercial housing and the average wage of on-post workers in the construction industry.

In terms of intergenerational structure, the proportion of construction-industry migrant workers aged 50 and above has undergone a leap-like increase. The nationwide migrant worker monitoring survey data for 2010 show this proportion at 12.1%. By 2024, it had risen to 31.6%—an increase of 19.5 percentage points over 14 years. The average age of construction-industry migrant workers has climbed in synchrony. Over the same period, the proportion of construction-industry workers among all migrant workers declined from approximately 16.1% in 2010 to 14.3% in 2024. The superimposition of a declining share and a rising age is not the independent effect of a single factor, but the projection in the data of a generational-exit pattern: "young people are no longer entering, and those who remain are continuing to age."

In terms of purchasing power, the average monthly income of construction-industry migrant workers over the same period grew from approximately 2,294 yuan in 2010 to approximately 7,460 yuan in 2024, an increase by a factor of roughly 2.25. The nationwide average sales price of commercial housing rose from 5,045 yuan per square meter in 2010 to 9,935 yuan per square meter in 2024, an increase by a factor of roughly 0.97. In terms of multiples, the growth rate of construction-industry wages outstripped the growth rate of housing prices. Yet the base price of housing far exceeds the monthly wage—in 2010, the monthly income amounted to 45.5% of the per-square-meter housing price; by 2024, it had risen to 75.1%. This ratio has risen, implying that absolute purchasing power has improved. However, the number of years of full salary required to purchase a home outright continues to exceed the effective expectation horizon of the individual worker. More importantly, after 2015, housing prices underwent a steep round of escalation, while wage growth remained relatively gentle, and the ratio underwent violent episodic deterioration during this period. This pattern of "overall improvement but severe episodic deterioration" carries the information that the narrative of "hard work brings prosperity" has suffered repeated frustrations in reality. Workers' incomes are rising, but every jump in housing prices has again pulled further away the distance to "purchasing a home and settling down on a construction-industry wage."

In terms of discourse, the rhetorical form of official pronouncements on "safety in production" has undergone a traceable shift. The dominant discourse in the years around 2000 was "Safety first, prevention foremost" and "Strictly prohibit operations in violation of regulations"—command-style, penalty-oriented. The language of the Ministry's annual safety-accident notifications was administrative: stating accident data, analyzing principal causes, setting forth rectification demands. After 2010, strongly normative formulations such as "Life supreme," "Red-line consciousness," and "Zero tolerance" began to appear with increasing frequency in official texts. The rhetoric of the notifications escalated from "An accident has occurred; we must draw lessons" to "Human life is of paramount value;

development must never be pursued at the cost of human life"—a shift from factual statement to **moral pledge**.This rhetorical escalation is not in itself a negative development—it reflects a genuine strengthening of regulatory intensity. Yet from the standpoint of discourse analysis, it simultaneously releases another signal: something that was originally supposed to be embedded within the operational logic of the industry now requires repeated, pointed, and high-decibel emphasis.

5.3.2 Trend Observations

The directions of movement of the three indicators are convergent. The labor-force structure reflects the behavioral judgment of construction workers—the young generation is making its choice by "no longer entering this industry." The purchasing-power ratio reflects **the material fulfillment of its promises**—wages are rising, but every jump in housing prices again pulls further away the distance to purchasing a home on a construction wage. The discursive shift reflects the reaction at the governance level: when an increasing number of workers no longer regard high-risk, heavy-manual-labor occupations as a reliable pathway to "prosperity through hard work," and when the very mechanism of labor-force reproduction in this industry begins to falter, "safety in production" ceases to be merely a technical regulatory problem and starts to be incorporated into the moral narrative of "people first" in order to regain legitimacy.

The three information streams point in the same direction: a kind of narrative that was once broadly taken for granted within the construction industry—that hard work improves one's life, that safety in production is merely a technical matter—is simultaneously loosening at every level. This loosening is not something that can be attributed to any single policy or any single year. Its persistence over the time series and its synchronicity across dimensions are what constitute its structural character.

Chapter 6 Discussion

6.1 What This Paper Has Accomplished

This paper has completed a proof of concept.

The three criteria of Complexity Metabolic Failure—which in the CMF paper remained at the stage of directional statements—have each, in this paper, undergone the transformation from "not directly measurable" to "operationalizable indicator." Criterion A has been transformed from "total coordination cost" into the long-term trend of standard revision frequency. Criterion B has been transformed from "crisis half-life" into a response latency demarcated by three key dates. Criterion C has been transformed from "explanatory coverage of the meta-narrative" into a cross-verification of three information streams: generational exit data, purchasing-power ratio attenuation, and the defensive turn in official-media discourse.

The common structure of these three transformations is logical reversal. When a grand-theoretical variable cannot itself be measured directly, one turns instead to measuring the system-level consequences produced by the absence of that variable. Coordination cost is not measurable—but the driver of coordination cost, the upward surge in rule-revision frequency, is measurable. Crisis half-life is not measurable—but the ordinal comparison of response latency with the first victim node is measurable. Meta-narrative coverage is not measurable—but the reverse signals that appear after coverage has been lost—the systemic exit of young labor, the attenuation of the purchasing-power ratio, the defensive escalation of official-media rhetoric—are measurable.

The empirical materials for all three transformations have been successfully obtained. The standard revision frequency exhibits a trend characterized by "initial stability followed by a surge," with representative standards displaying revision intervals of over a decade to more than two decades. The response latencies for the two accident types are approximately 13 and 15 years, respectively. The proportion of construction-industry migrant workers aged 50 and above has increased by nearly twenty percentage points over 14 years; the purchasing-power ratio has undergone severe episodic deterioration; and the official discourse on safety in production has escalated from administrative factual statement to moralizing, repeated proclamation. All these materials derive from publicly available government documents and official statistical platforms. Any researcher with basic academic retrieval skills can independently replicate this acquisition process.

This paper has demonstrated, under resource constraints, that the pathway from the amorphous variables of grand theory, from directional statements to operationalizable indicators, exists and is traversable. This is the core contribution of this paper.

6.2 A Methodological Statement on Resource Limitations

All the empirical materials in this paper come from publicly accessible official data sources—the annual reports of the National Bureau of Statistics, the proclamations of the Ministry of Housing and Urban-Rural Development, the National Standard Information Service Platform. There is no access to non-public administrative databases, no commercial datasets, no division of labor within a research team. This is work completed within the standard resource boundaries of an independent researcher.

The resource limitations are not a flaw of this paper—they are its premise. What this paper seeks to demonstrate is precisely that, even under the most constrained resource conditions, even without a team, without funding, without data privileges, the pathway of operationalizing grand-theoretical variables can still be traversed. If this pathway can be walked with the narrowest of channels, then under more generous resource conditions, it can certainly be walked further.

To openly acknowledge this is both a matter of academic honesty and a methodological encouragement. It means that the operationalization protocol presented in this paper is replicable for any researcher situated under similar resource constraints. Resource limitations are not a shackle on operationalization—they merely demarcate an executable boundary within which operationalization must land. Within that boundary, the pathway is open.

6.3 What This Paper Has Not Accomplished

What this paper has not done must be stated as clearly as what it has done.

This paper has not calibrated failure thresholds. At what point does the revision frequency of Criterion A tip from "normal" to "out of control"? How long a latency counts as "deadlock" for Criterion B? How high a proportion of generational exit counts as "irreversible attenuation" for Criterion C? None of these questions are answered in this paper. The calibration of thresholds requires the accumulation of multiple cases, iterative parameter adjustment, and cross-sectoral comparison—it is an independent systems-engineering project. This paper has only obtained the ruler; it has not yet inscribed the graduations upon it.

This paper has not performed cross-criterion coupling and integration. The three criteria were each independently operationalized and each independently exhibited trends, but they have not yet been combined into a synthetic judgment model. Whether there exist sequential triggering relationships among the frequency surge of Criterion A, the response latency of Criterion B, and the generational attenuation of Criterion C—and whether one criterion plays a dominant role—these coupling questions are not addressed in this paper.

This paper has not generalized its conclusions to industries beyond the construction sector. The construction industry was selected as the proving ground because its data openness, regulatory completeness, and the accessibility of sectoral cognition are optimal under the resource constraints of an independent researcher. Whether other industries—finance, environmental protection, education, healthcare—also possess operationalizable criterion proxy variables is not asserted by this paper. That is a task for subsequent extension studies.

This paper has not performed a deviation-degree calculation. A deviation-degree calculation requires an independent scholarly baseline as a reference, yet the three CMF criteria, as newly operationalized criteria undergoing their first calibration, have not yet accumulated a repository of widely-discussed and independently verified baselines in the academic literature. The endpoint of this paper lies at the point where criterion operationalization and calibration is completed. Subsequent researchers may, on the basis of the criteria calibrated in this paper, after selecting specific historical cases and confirming usable independent baselines, directly feed these criteria into the Boundary Anchoring Method to calculate deviation degrees and complete a quantitative location of system states.

These instances of "not accomplished" are not flaws inherent to this paper. The task of a proof-of-concept paper is not to exhaust all subsequent work, but to demonstrate that the premise of that subsequent work holds. If the criteria cannot be operationalized, the calibration of thresholds cannot even begin. If proxy variables cannot be obtained, there are no materials with which to perform coupling and integration. This paper has demonstrated that premise—the criteria can advance from directional statements to operationalizable indicators, and proxy variables can be obtained from publicly available data. The starting point of the subsequent work is precisely the endpoint of this paper.

6.4 The "Representative" Status and "Open-Interface" Character of This Paper

Complexity Metabolic Failure is not the only grand-theoretical concept in need of operationalization. Marx's "class consciousness," Weber's "charisma," Durkheim's "social solidarity," Foucault's "dispersion of power"—every grand-theoretical variable confronts the same predicament: the concept is rigorously coherent within its theoretical architecture, yet finds no direct counterpart in the empirical world.

This paper selects the CMF criteria as a case-concept and executes a complete demonstration of operationalization. The coverage of Criterion A (efficiency layer), Criterion B (temporal layer), and Criterion C (meaning layer) in terms of variable types gives this demonstration a relevance to the shared predicament of grand theory that extends beyond the single case. It is not a method tailored to one specific concept, but a transplantable transformation logic—identify, through the original predicament, why the variable cannot be measured; execute, through the key reversal, the displacement of the measurement object from the variable itself to the consequences of its absence; declare, through the preconditions, the presuppositions on which the reversal rests; and lock in, through the operationalization result, the observable signal. The structure of this four-step logical chain does not depend on the specific content of CMF theory. Any grand-theoretical variable, so long as the absence of that variable would leave observable traces at some layer of the system, can attempt to enter the same logical reversal chain.

This paper is therefore not merely a case paper that "presents the CMF criteria using data from the construction industry." It is simultaneously an open interface. What it provides is not a conclusion, but a method—not the fixed pairing of "this variable can be proxied by this indicator," but the thinking procedure of "how to derive an observable signal from an amorphous variable." Subsequent researchers who attempt to operationalize other grand-theoretical variables can directly adopt the four-step logical chain framework of Chapter 3 of this paper, treating "original predicament → key reversal → preconditions → operationalization result" as a template into which new content can be filled. If successful, the method-repository for the operationalization of grand theory gains a new case. If unsuccessful, the failure will occur at a specific step in the logical chain—was the predicament misidentified? Was the reversal direction off? Did the preconditions not hold? Did the operationalization result not match? Failure becomes traceable, correctable, and convertible from an individual's trial-and-error into experience that subsequent researchers can draw upon.

This paper declared its identity in the introduction—proof of concept, not theory confirmation. This chapter reaffirms the boundaries of that identity: this paper is the first segment of the bridge. The pathway is now open, and the endpoint lies at the next step. The research that follows can, from this starting point, turn to the next tasks—calibrating the thresholds one by one, attempting cross-criterion coupling and integration, and testing the transplantability of criterion operationalization in industries beyond the construction sector. These tasks no longer fall within the scope of a proof of concept. They belong to the research agenda that naturally unfolds after a proof of concept has succeeded.

Chapter 7 Conclusion

Grand social theory has long confronted an awkward predicament: the most profound theories are often those that employ the least measurable variables. This paper has sought to respond to that predicament.

Taking the three criteria of Complexity Metabolic Failure as a case instance, this paper has demonstrated that the transformation pathway from directional statements to operationalizable indicators exists and is traversable. Criterion A has been transformed from "total coordination cost" into the upward-surge trend of standard revision frequency. Criterion B has been transformed from "crisis half-life" into a response latency demarcated by three key dates. Criterion C has been transformed from "explanatory coverage of the meta-narrative" into a cross-verification of three information streams: generational rupture, purchasing-power attenuation, and discursive shift. The common structure of these three transformations is logical reversal—when the variable itself cannot be measured, one turns instead to measuring the system-level consequences produced by the absence of that variable.

This paper does not claim that failure is currently occurring—the thresholds have not yet been calibrated, and this judgment cannot be made. This paper has demonstrated only that the amorphous variables of grand theory can advance from directional statements to operationalizable indicators, and that under the resource constraints of an independent researcher, this transformation protocol is fully executable. The empirical materials have been successfully obtained from publicly available official data, and the direction of the trends is broadly consistent with theoretical expectations. The pathway is now open.

This paper is the first brick laid upon this pathway. Subsequent researchers may, on the basis of the criteria calibrated in this paper, continue to advance the tasks of threshold calibration, criterion coupling, and cross-sectoral extension. This paper is also a complete operational exhibition of the Criterion Dual-Reversal Method—the step-by-step derivation of observable signals from abstract pathological concepts, with the reversal logic of each step laid transparently open for subsequent researchers to inspect. If any link in the logical reversal chain presented in this paper is proven mistaken, the error can be located and corrected—this is precisely the value of transparent argumentation: it permits itself to be surpassed.

The empiricalization of grand theory is not a task that can be accomplished by a single paper. The significance of this paper does not lie in how far it has traveled, but in having demonstrated that a pathway long tacitly assumed to be closed is, in fact, traversable. It is not the end of the road; it is only the first segment. The road is now open. The next researcher can continue further down this road.

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